Thaer Baniatta

Electrical Engineer · VLSI Analog & Mixed-Signal Layout

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PROFILE SUMMARY

Electrical Engineer with 2 years of professional experience in VLSI analog and mixed-signal layout design, specializing in physical layout implementation, floorplanning, device placement, routing, and physical verification (DRC/LVS).

Experienced in debugging layout connectivity issues, improving device matching, and considering parasitic effects to support reliable integrated circuit designs.

Also experienced in software development, embedded systems, and programming, with a growing interest in design automation, Tcl scripting, and semiconductor physical integration workflows.

EXPERIENCE

VLSI Analog & Mixed-Signal Layout Engineer — Golden Electronics · 2 years

- Implemented physical layout for analog and mixed-signal designs, covering floorplanning, device placement and routing.
- Performed physical verification with DRC and LVS checks.
- Debugged layout connectivity issues.
- Improved device matching and accounted for parasitic effects to support reliable integrated circuits.

Content Moderator — CSmena · 5 months

Reviewed user-generated content against platform policies to keep the service safe and compliant.

EDUCATION

B.Sc. in Electrical Engineering — Graduated end of 2024

Jordan University of Science and Technology (JUST), Irbid, Jordan

Major courses: Digital Logic Design and Computer Architecture, Microprocessor Systems, Digital Electronic Circuits, Electronic Circuits, Control Systems, Digital Control, Communication Systems, Signals and Systems Analysis, Power Electronics, Electric Machines, Measurement Systems and Sensors.

Graduation project: A smart home system that connects sensors and actuators for home automation.

42 Amman — In progress

Software development program (C and C++, Linux system programming), affiliated with CPF.

TECHNICAL SKILLS

IC layout	Analog and mixed-signal layout, floorplanning, device placement, routing, DRC/LVS, matching, parasitic awareness
Programming	C, C++, Tcl scripting for design automation
Tools	Custom Compiler, Jira, Linux, Git, Makefiles
Systems	Sockets, processes and pipes, multithreading
Working style	Problem solving, teamwork, critical thinking

PORTFOLIO — SOFTWARE PROJECTS (GITHUB.COM/THAQER)

webserv (C++98, team of 3): HTTP/1.1 server built from scratch around a single poll() loop, with virtual servers, uploads, directory listing, redirects, non-blocking CGI and chunked requests.

minishell (C, team of 2): Bash-like shell with a tokenizer, AST parser, expander and executor; pipes, redirections, heredocs, builtins and signal handling.

philosophers (C): dining philosophers with pthreads and mutexes plus a monitor thread; clean exit and no Valgrind errors in tested runs.

Inception (Docker Compose): WordPress stack of three custom images (NGINX with TLS, WordPress with PHP-FPM, MariaDB) with Compose secrets, one network and persistent volumes.

Also: push_swap (100 numbers in 1,084 moves), pipex, so_long, and C++ Modules 00 to 09 (templates, STL, polymorphism).

TRAINING

- IEEE courses on Arduino and PCB design.
- IEEE workshops on electric cars and fixed-wing drone manufacturing.

LANGUAGES

Arabic (native) · English (very good)